(d) Explain how can the process is changing its size. 7

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B. E. (Eighth Semester) Examination, April-May 2016

(Old Scheme)

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(IT Engg. Branch)

DESIGN of UNIX OPERATING SYSTEM

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all question. Sub question (a) of 2 marks is compulsory, Attempt any two sub question of 7 marks from (b), (c) and (d).

Unit-I

1. (a) What do you mean by multiuser system. 2

(b) Explain the following commands with their syntax and usage. 7

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(i) Man

(ii) head

(iii) cat

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(iv) bc,

(v) wc,

(vi) grep,

(vii) sed

(c) Explain structure of UNIX OS write features of Unix OS. 7

(d) Mention all modes of Vi editor and also explain what can be done in each of these modes. 7

Unit-II

2. (a) What do you mean by shell variable? 2

(b) Write a shell script to generate the factorial of a given number through keyboard. 7

(c) Explain AWK pattern scanning and processing. Also explain what are the advantages of AWK scanning and processing. 7

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(d) Explain arrays in AWK and also given suitable example of AWK array. 7

Unit-III

3. (a) Define buffer header. 2

(b) Explain the architecture of UNIX operating system with the block diagram. 7

(c) List all the advantages and disadvantages of Buffer cache. 7

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(d) Explain all scenarios of buffer retrival. 7

Unit-IV

4. (a) Define INODES. 2

(b) How a new inode number is allocated to a file. Explain iget algorithm. 7

(c) Explain namei algorithm in detail. 7

(d) Explain the algorithm for creating a file. 7

Unit-V

5. (a) Define process in Unix O.S. . 2

(b) What is the context of a process and what are its various components of context. 7

(c) Explain the process states and transition layout of the unix O.S. . 7

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